Oluwatosin Ayeni

Staff Cluster Infrastructure Engineer — Anthropic (London)

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PROFILE

Staff-calibre ML Infrastructure Engineer with hands-on experience designing and operating production GPU clusters, Kubernetes serving stacks, and observability infrastructure at scale. Built a full GKE GPU cluster (NVIDIA L4 24GB) with Terraform IaC, KEDA autoscaling on GPU queue depth, 27-target Prometheus + DCGM stack, Argo CD GitOps, NetworkPolicy default-deny, and Kyverno pod security — serving 723 live requests and 1.25M+ tokens over 5.3 days. CUDA-level hardware profiling on H100 NVL (Nsight Systems/Compute) achieving 1,443× TTFT speedup. Committed to building the reliable infrastructure that makes Claude possible.

CORE COMPETENCIES

Kubernetes & Cluster Ops	GKE, HPA/KEDA autoscaling (GPU queue depth), Argo CD GitOps, Argo Rollouts canary, NetworkPolicy default-deny, Kyverno pod security, Velero backup, namespace resource quotas
Infrastructure as Code	Terraform (GKE cluster, node pools, networking, IAM, GCS), Helm charts, GitHub Actions CI/CD, declarative GitOps pipelines
GPU Infrastructure	NVIDIA L4/H100 NVL production experience, DCGM Exporter, BFloat16/AWQ quantisation, CUDA-level memory profiling, Nsight Systems/Compute, NVLink topology awareness
Observability	27-target Prometheus stack, Grafana dashboards, AlertManager (6 rules), Loki + FluentBit DaemonSet, KEDA queue-depth metrics, DCGM GPU telemetry
Networking & Security	Traefik TLS ingress, NetworkPolicy default-deny, cert-manager, Kyverno admission control, least-privilege RBAC, mTLS service mesh patterns

PROFESSIONAL EXPERIENCE

Jan 2025–present — **Founder & Lead Engineer** — *Nexora Academic*, Glasgow, UK

- H100 NVL benchmark: vLLM v0.30 vs SGLang v0.5.20 — Qwen3-8B BF16, TP=2. Peak: 5,965 tok/s; RadixAttention: 1,443× TTFT speedup (39 ms vs 55,565 ms) at 131K-token context; 100% SLO compliance up to concurrency 128. Live: inference-web-theta.vercel.app
- Built *inference-optimizer*: production AI Inference Gateway — 3.87× faster (parallel execution), 61.4% cost reduction (prompt caching), 14× throughput (concurrency), 64.3% cheaper (model routing). 13 optimisations, 4 providers.
- Built *vllm-k8s-stack*: Kubernetes LLM serving — KV-cache-aware routing, HPA autoscaling on GPU pools, Prometheus/Grafana observability.
- Built *inference-harness*: unified benchmark framework spanning closed APIs + vLLM/SGLang/TensorRT-LLM with GPU profiling and live dashboard.
- Designed 13-phase AI Inference Engineering curriculum covering GPU architecture, CUDA optimisation, serving frameworks, and distributed inference.

Jan 2023–present — **Founder & AI Consultant** — *Vertinab / AnaburyAI Technology*, Glasgow, UK

- 28× LLM cost reduction deploying Qwen2.5-7B on RTX 4090 via vLLM/SGLang with continuous batching and KV cache optimisation.
- CUDA KV cache kernel profiling with NVIDIA Nsight: identified memory bandwidth bottleneck; 3.8× throughput improvement; 18× latency reduction via continuous batching redesign.
- Fine-tuned Mistral 7B with LoRA/PEFT on radiology corpus: +22% diagnostic accuracy, −35% hallucination rate; deployed as production clinical inference API.

2024–present — **Senior Software Engineer (Contract)** — *EduTechConnect / Stellar Labs*, Remote

- Real-time NLP pipeline (WhatsApp → Zoho CRM → Supabase) with sub-500 ms end-to-end latency under concurrent load.

Mar 2026–present **AI Automation Engineer** — *EduTech Global*, Remote

- LLM-backed student chatbots with structured output extraction and multi-step reasoning for Ahmadu Bello and Babcock universities.

2018–2022 **Data Analyst** — *Bytelab Technologies Ltd*, Nigeria

KEY PROJECTS

llm-inference-platform (GKE) Full production GPU cluster: Terraform IaC, NVIDIA L4, KEDA queue-depth autoscaling, Argo CD GitOps, Argo Rollouts canary, Traefik TLS, NetworkPolicy default-deny, Kyverno pod security, 27-target Prometheus + Loki. LIVE: 723 req, 1.25M+ tokens. GitHub

llm-inference-benchmark (H100) CUDA-level profiling on 2× H100 NVL 94 GB (NVLink 4.0, CUDA 13.4). Nsight Systems/Compute kernel analysis. 1,443× TTFT speedup. 5,965 tok/s peak. Live results

vllm-k8s-stack Kubernetes LLM serving: KV-cache-aware routing, HPA/KEDA autoscaling, hostPath weight caching (5 min cold → 30 s cached), Prometheus/Grafana/AlertManager, Argo CD GitOps. GitHub

inference-optimizer Production AI Gateway: 13 optimisations, multi-provider (Claude, GPT, Gemini), 3.87× faster, 14× throughput, 61.4% cost reduction. GitHub

open-model-engine GPU-first infrastructure benchmarking: vLLM, SGLang, TensorRT-LLM, CUDA profiling, quantisation, speculative decoding. GitHub

EDUCATION

2022–2023 **MSc Artificial Intelligence and Applications, Distinction** — *University of Strathclyde*, Glasgow, UK

2010–2016 **MTech / BSc Physics** — *Federal University of Technology Akure*, Nigeria

OPEN SOURCE

GitHub 43 public repos — github.com/AyeniOluwatosinOlawale